

Regularisation and early stopping

Project: Image-pattern Network Investigation

1. Your mission

Your learning target

I can separate a weight penalty from prediction error.

I can trace a regularised update under a stated objective.

I can apply a stopping rule and restore the best observed checkpoint.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Training and validation curves. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Batching and epochs. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Gradients with a numerical check. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Separate a weight penalty from prediction error

Trace a regularised update under a stated objective

Apply a stopping rule and restore the best observed checkpoint

Retrieve what you already know

If validation chooses an earlier checkpoint, which parameters should be used after selection?

My starting explanation

This week's end goal

By the end, I can explain regularisation and early stopping using a traced example and a stated limitation.

2. Notice the evidence

Keep $h = \text{ReLU}(x)$, $y = vh$, train only v on $(1, 2)$ and $(2, 4)$, and freeze all hidden parameters and the zero output bias. Training MSE is $L(v) = 2.5(v - 2)^2$. Define $J(v) = L(v) + \lambda v^2$, with λ nonnegative. Only this one output weight is penalised here.

For $\lambda = 0.5$, the penalty at $v = 1.5$ is $0.5 \times 2.25 = 1.125$. Prediction MSE is 0.625 , so total objective J is 1.75 . Keep these quantities separate in logs. A smaller total objective does not automatically mean a smaller prediction error.

v · Prediction MSE · Penalty at lambda 0.5 · Total J

v: 1

Prediction MSE: 2.5

Penalty at lambda 0.5: 0.5

Total J: 3

v: 1.5

Prediction MSE: 0.625

Penalty at lambda 0.5: 1.125

Total J: 1.75

v: 2

Prediction MSE: 0

Penalty at lambda 0.5: 2

Total J: 2

Among these three candidates, the smallest penalised objective is at $v = 1.5$, whereas the smallest training MSE is at $v = 2$. Penalising a weight changes the tradeoff. It does not guarantee better validation

performance or fair outcomes.

List the given facts and one fact that is not supplied.

3. See how it works

The data gradient is $5(v - 2)$. The penalty gradient is 2 times lambda times v , giving total gradient $5(v - 2) + 2 \text{ lambda } v$. At $v = 1.5$ and $\text{lambda} = 0.5$, these contributions are -2.5 and 1.5 , so total gradient is -1 . A rate 0.1 step gives $v = 1.6$.

```
def terms(v, penalty):  
    data_loss = 2.5 * (v - 2) ** 2  
    weight_cost = penalty * v * v  
    gradient = 5 * (v - 2) + 2 * penalty * v  
    return data_loss, weight_cost, data_loss + weight_cost, gradient  
  
v = 1.5  
penalty = 0.5  
old = terms(v, penalty)  
v = v - 0.1 * old[3]  
print(round(v, 6))  # 1.6  
print([round(x, 6) for x in terms(v, penalty)])  
# [0.4, 1.28, 1.68, -0.4]
```

Assume finite numeric values and nonnegative penalty. At the new v , data loss is 0.4 , penalty 1.28 and total 1.68 . Choose penalty strength using a declared validation prediction metric, not by comparing total objectives with different lambdas as if they were the same measure.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Compare a familiar case with a changed case

Start $v = 1.6$ with $\lambda = 0.5$ and rate 0.1. Calculate both gradient contributions, new v and new prediction MSE.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change: Change patience from 1 to 2 under the same tie rule.

Prediction: Stop at checkpoint 4; still restore checkpoint 2.

Reason: Checkpoints 3 and 4 are two consecutive non-improvements; a later stop does not redefine the best checkpoint.

5. Words that unlock the idea

Word · Meaning

Word: Regularisation

Meaning: A constraint or training procedure intended to limit fitting choices and improve generalisation

Word: Weight penalty

Meaning: An added objective term that discourages specified large parameter values

Word: Patience

Meaning: The allowed run of consecutive non-improving evaluations before stopping under a stated rule

Word: Restore

Meaning: Reload the parameters saved at the selected checkpoint

Use two terms to explain a step in this week's example.

6. Follow a worked example

Patience two with strict improvement

Epoch · Validation MSE · Decision

Epoch: 0 then 1 then 2

Validation MSE: 4 then 2 then 1.5

Decision: Save each improved state

Epoch: 3 then 4

Validation MSE: 1.6 then 1.4

Decision: Wait once then reset

Epoch: 5 then 6

Validation MSE: 1.45 then 1.46

Decision: Two misses so stop at 6

Initial checkpoint is epoch 0. Evaluate once after each epoch. A strict decrease is improvement; an exact tie is a miss. Reset the miss count after improvement and stop at two consecutive misses. Restore epoch 4, the best observed checkpoint.

At epochs 3, 4, 5 and 6, record the miss count after evaluation. Which epoch stops training, and which epoch supplies the restored parameters?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Regularisation and early stopping

Separate a regularisation penalty from prediction loss and an early-stopping rule.

The facts we will use

Toy objective $J(a)=(a-3)^2+2a^2$; start $a = 0$; gradient $6a-6$; rate $1/6$. Separate validation losses: $[5, 3, 3, 4]$, patience 1, no ties as improvement.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

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Regularisation and early stopping

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Regularised update

At 0 gradient -6; update gives $a = 1$.



2. Split objective

At 1 prediction term 4, penalty 2, total 6; at 0 total 9.



3. Early-stop separately

Best validation checkpoint is 2 with loss 3; checkpoint 3 is no improvement, so stop at 3.

CHANGE ONE THING

Change patience from 1 to 2 under the same tie rule.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Regularised update

What happens and why: At 0 gradient -6; update gives $a = 1$.

Step: 2. Split objective

What happens and why: At 1 prediction term 4, penalty 2, total 6; at 0 total 9.

Step: 3. Early-stop separately

What happens and why: Best validation checkpoint is 2 with loss 3; checkpoint 3 is no improvement, so stop at 3.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Regularisation adds a parameter penalty to the training objective. Here it discourages large a , so the objective balances prediction error against that penalty. Early stopping is a different decision based on the supplied validation sequence. A tied validation loss is not improvement under this policy; stop when the patience allowance is used and restore the best checkpoint.

What this example does not establish

Regularisation and early stopping are different tools; neither guarantees better performance in every dataset.

7. Find and repair a mistake

Early stopping limits the training duration using development evidence; it does not add a weight term to the objective. For this policy, initialise the best checkpoint to epoch 0 and the miss count to zero. After each later evaluation, save and reset only on a strict improvement; otherwise increment the count. Stop as soon as the count reaches patience, then reload the best checkpoint.

Suppose a pre-recorded demonstration also lists epoch 7 loss 1.3. A live run following the rule has already stopped at 6 and does not observe 7. Do not use future values to rewrite the stopping decision. The policy can miss a later improvement; patience balances waiting against a training budget and must be specified in advance.

Save actual model parameters with the checkpoint, not just its epoch number or loss. With a fixed

maximum epoch budget, stop when either limit is reached and restore the best observed state. An exact tie keeps the earlier checkpoint under our rule. Different minimum-improvement thresholds are possible, but would be different declared policies.

Locate the earliest unsupported step

Validation MSE at epochs 0 to 7 is [4, 2, 1.5, 1.6, 1.4, 1.45, 1.46, 1.3]. Apply the lesson policy with patience 2. Give the stop epoch, restored epoch and why the listed epoch 7 cannot change this decision.

State one repair and the checks you would rerun.

8. Practise together

At $v = 2$ and $\lambda = 0.5$ in the lesson, compute prediction MSE, penalty, total objective and total gradient. Does zero training MSE make this gradient zero?

At epochs 3, 4, 5 and 6, record the miss count after evaluation. Which epoch stops training, and which epoch supplies the restored parameters?

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the next update

Use workbook W2 for the connected example “Regularisation and early stopping”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Regularisation and early stopping. Proposed explanation: “Zero prediction error means no regularised update”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why should the penalty value not be reported as validation error? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

Predict, investigate and preserve the evidence

Start $v = 1.6$ with $\lambda = 0.5$ and rate 0.1. Calculate both gradient contributions, new v and new prediction MSE.

```
def terms(v, penalty):
    data_loss = 2.5 * (v - 2) ** 2
    weight_cost = penalty * v * v
    gradient = 5 * (v - 2) + 2 * penalty * v
    return data_loss, weight_cost, data_loss + weight_cost, gradient

v = 1.5
penalty = 0.5
old = terms(v, penalty)
v = v - 0.1 * old[3]
print(round(v, 6))  # 1.6
print([round(x, 6) for x in terms(v, penalty)])
# [0.4, 1.28, 1.68, -0.4]
```

Record · Your evidence

Record: Given input and fixed rule

Your evidence: _____

Record: Prediction before checking

Your evidence: _____

Record: Actual result, or “not run”

Your evidence: _____

Record: First mismatch and one explanation

Your evidence: _____

Record: Boundary or missing input and expected response

Your evidence: _____

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

Explain → apply → challenge

At $v = 2$ and $\lambda = 0.5$ in the lesson, compute prediction MSE, penalty, total objective and total gradient. Does zero training MSE make this gradient zero?

Validation MSE at epochs 0 to 7 is [4, 2, 1.5, 1.6, 1.4, 1.45, 1.46, 1.3]. Apply the lesson policy with patience 2. Give the stop epoch, restored epoch and why the listed epoch 7 cannot change this decision.

A team compares $\lambda = 0$ and $\lambda = 0.5$ by choosing whichever has the smaller penalised training objective. Explain why this does not establish the better prediction model and name a suitable development comparison.

Explain which evidence supports your conclusion and which claim remains unproved.

11. Connect your project

Add a labelled regularisation and early stopping evidence sheet to Image-pattern Network Investigation. Show how you can separate a weight penalty from prediction error. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence · Record

Project evidence: Component and version

Record: _____

Project evidence: Inputs permitted at prediction time

Record: _____

Project evidence: Trace and comparison with earlier work

Record: _____

Project evidence: Failure, limit and human review

Record: _____

Evidence for your project

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

Weight penalties and early stopping can be used together, but they act differently. Record which weights receive penalties, the objective convention, lambda, learning rate, checkpoint frequency, patience, tie rule and maximum budget. For this classroom model there are no extra learned preprocessing or running statistics; larger models may require more state to reproduce evaluation.

A team stops at epoch 6, uses its final parameters instead of the saved best state, and chooses patience by looking at final test losses. Identify the two mistakes and explain how to repair the experiment.

Select these settings using validation, freeze the complete procedure, and reserve untouched final test data for reporting. Neither regularisation nor stopping makes a weak validation set representative. Next week studies initialisation and reproducibility.

A team compares lambda 0 and lambda 0.5 by choosing whichever has the smaller penalised training objective. Explain why this does not establish the better prediction model and name a suitable development comparison.

My limited claim and a useful next check

13. Study a complete case

Complete evidence record

Patience two with strict improvement

Epoch · Validation MSE · Decision

Epoch: 0 then 1 then 2

Validation MSE: 4 then 2 then 1.5

Decision: Save each improved state

Epoch: 3 then 4

Validation MSE: 1.6 then 1.4

Decision: Wait once then reset

Epoch: 5 then 6

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Decision: Two misses so stop at 6

Initial checkpoint is epoch 0. Evaluate once after each epoch. A strict decrease is improvement; an exact tie is a miss. Reset the miss count after improvement and stop at two consecutive misses. Restore epoch 4, the best observed checkpoint.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

14. Try a new case independently

Independent case: Transfer both rules independently

For a new objective $J(a) = (a - 2)^2 + a^2$, gradient $4a - 4$, start $a = 0$ and take one step at rate 0.25. Give new a , prediction error term, penalty and total J . Separately apply patience 2 with strict improvement to validation values $[3, 2, 2, 2, 1]$ at epochs 0 to 4; report the stop and restored epochs.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

For a new objective $J(a) = (a - 2)^2 + a^2$, gradient $4a - 4$, start $a = 0$ and take one step at rate 0.25. Give new a , prediction error term, penalty and total J . Separately apply patience 2 with strict improvement to validation values $[3, 2, 2, 2, 1]$ at epochs 0 to 4; report the stop and restored epochs.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.